

PASINDU WIDUSHAN

Engineering Undergraduate

Software Engineer | AI & Machine Learning | Building AI Powered Solutions | Python | Full Stack Dev

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| <https://github.com/widushan> |

| <https://widushan.github.io/My-Portfolio/> |

PROFILE

Motivated undergraduate specializing in advanced technologies including Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and AI-driven automation. Demonstrates strong technical competence in Python, Java, PHP and MERN stack web development, and with ongoing development in workflow automation. Experienced in designing and implementing innovative academic and personal projects with a focus on efficiency, maintainability, and real-world impact. Dedicated to pursuing a career as a Software Engineer – AI ML , Full Stack with a commitment to continuous learning, innovation, and professional excellence.

EXPERIENCE

Assistant Accountant

Waguruwela Oil Mills (Pvt) Ltd | 2021 - 2022

Mathematics Tutor (Private)

2019 – Present

EDUCATION

BSc (Hons) Eng. in Information and Communication Engineering

SLTC Research University | 2022 – 2026

Specialization in Telecommunication. Coursework: programming, software development, electronics, communication systems, data analytics, embedded technologies, Project planning.

Advanced Level – Physical Science Stream

Saranath Natinal College | 2020

Combined Maths: C, Chemistry: C, Physics: C

General English: A

SKILLS

- Programming & Software Development – Python, JAVA, PHP
- AI / ML / DL – TensorFlow, scikit-learn, Deep Learning Models, Computer Vision, Model Training & Evaluation
- Generative AI & AI Agents – LLMs, NLP, LangChain, CrewAI
- Web & App Development – MERN, Next.js, Flutter, Django, Laravel
- Databases – SQL, NoSQL, MongoDB, Vector DBs
- Other Skills – Git, API Integration, Problem Solving & Debugging, Agile, DevOps Basics

COURSES

- **Professional Certificate in Machine Learning**
Informatics Institute of Technology — 2025
- **AI/ML Engineer – Stage 1 & Stage 2**
Sri Lanka Institute of Information Technology — 2024
- **Machine Learning & AI/ML Projects**
Simplilearn — 2025
- **Python for Beginners , Python Programming**
University of Moratuwa — 2023
- **Web Design for Beginners , Front-End Web Development**
University of Moratuwa — 2023
- **AWS Academy Graduate – Cloud Foundations**
Amazon Web Services — 2025

PROJECTS

1. Personal AI Assistant

A Real-Life Personal AI Assistant | Python

Trained a multi-modal Personal AI Assistant in Python featuring real-time voice recognition, text-to-speech, web search, AI image generation, and task automation. Produced with PyQt5 for a modern GUI, it supports interactive chat and voice commands for seamless user interaction.

- Core Technologies: Python, PyQt5, Speech Recognition, pyttsx3, OpenAI API, Requests, python-dotenv, Real-time Web Search, AI Image Generation, Task Automation

2. RAG Chat Bot

RAG Chatbot with JavaScript, LangChain.js, Next.js, OpenAI

Programmed a Retrieval-Augmented Generation chatbot using Next.js, LangChain.js, and OpenAI. It combines document retrieval with AI generation to provide intelligent, context-aware responses. The app features a modern, responsive UI, real-time chat, and intelligent document parsing for enhanced user interactions.

- Core Technologies: Next.js 14, React, TypeScript, Tailwind CSS, LangChain.js, OpenAI API, Vector Database

3. News Research Tool

End to End Gen AI Project Using Langchain, OpenAI

Derived a News Research Tool leveraging LangChain, OpenAI, and FAISS to transform how users interact with news articles. Users can input multiple URLs, ask questions, and receive AI-powered answers with source citations, all through a clean Streamlit web interface.

- Technologies: Python, Streamlit, LangChain, OpenAI GPT, FAISS (Vector Search)

4. AI Career Coach

AI-Powered Full-Stack Career Guidance Platform | Next.js, LLM, AI Built a production-ready AI Career Coach platform with resume analysis, AI cover letter generation, career roadmaps, and real-time mentorship chat using LLM-powered agents and scalable full-stack architecture.

- Core Technologies: Next.js 15, React 19, TypeScript, Tailwind CSS, LangChain, OpenAI API, Inngest (Agent Workflows), Neon PostgreSQL, Drizzle ORM, Clerk Authentication, Supabase, Vector Database, Vercel Deployment.

<https://educational-ai-agent-xi.vercel.app/>

5. Medical AI Voice Agent

AI-Powered Voice Consultation Platform | LLM + Voice AI

Built a voice-enabled AI medical consultation system integrating real time AI voice calls, live transcript streaming, and GPT power medical report generation. Designed multi-agent workflows for specialist consultations with persistent session history and secure authentication.

- Tools & Technologies: Next.js 15, React 19, TypeScript, OpenAI API (GPT-4o-mini), Vapi AI, Neon PostgreSQL, Drizzle ORM, Tailwind CSS, Clerk Authentication, Vercel
<https://medical-ai-voice-agent-two.vercel.app/>

6. AI Travel Planner

AI-Powered Full-Stack Travel Planning Application | React, Generative AI

Developed an AI-powered travel planner that generates personalized day-by-day itineraries using user inputs, integrating real-time location services, Google authentication, cloud storage, and dynamic content for a seamless, interactive experience.

- Core Technologies: React 19 (Vite), Tailwind CSS, shadcn/ui, Firebase Firestore, Google Places API, Google OAuth, Generative AI APIs, React Router DOM, Vercel.
<https://ai-travel-planner-nine-nu.vercel.app/>

7. Heart Disease Prediction System

A Machine Learning-powered web application

Developed a Machine Learning-based Heart Disease Prediction System with a Flask web app, using a trained Random Forest model and real-time preprocessing to deliver accurate, user-friendly risk predictions.

- Tools & Technologies: Python, Flask, scikit-learn, Random Forest, NumPy, Pandas, HTML/ CSS, Pickle

8. Digital Facial Biomarkers for Age Related Health Condition Detection Parkinson's Disease

Deep Learning model with real time web screening

Built a multimodal AI framework detecting Parkinson's Disease from facial videos by fusing OpenFace-extracted Action Units with PubMedBERT-encoded clinical text through cross-attention, using GCN, DDCA and CDCA. Deployed as a web app for webcam/upload-based screening.

Tools & Technologies: Python, OpenFace, PubMedBERT, GPT-4o, FFmpeg, HTML, CSS, JavaScript

9. Stock Trend Prediction

Predicting Stock Market Trends with Deep Learning

Created a deep learning-based Stock Trend Prediction system using LSTM networks to forecast stock prices and analyze market trends. Crafted a Flask web app for real-time stock data analysis, interactive EMA charts, and downloadable datasets.

- Tools & Technologies: Python, Flask, TensorFlow/ Keras, LSTM Neural Networks, Pandas, NumPy, Scikit-learn, yfinance, Matplotlib, Plotly, Bootstrap 5

10. E-commerce Website

Full Stack E-commerce Website Using React JS | MERN Stack

eCommerce Project Launched a full-stack e-commerce platform for fashion and lifestyle products. Features include user authentication, admin panel, product catalog, shopping cart, checkout, and order management, all incorporated with React, Node.js, Express, and MongoDB.

- Tech Stack: React, Vite, Tailwind CSS, Node.js, Express, MongoDB, Cloudinary, Vercel
<https://trendzide-frontend.vercel.app/>

11. LMS Website

Full Stack Learning Management System (MERN)

Developed a production-ready Learning Management System with secure Authentication, role-based access control, Stripe payment integration and educator dashboards. Implemented course enrollment workflows, progress tracking, protected REST APIs, and webhook-based payment confirmation for a scalable EdTech Platform.

- Tools & Technologies: MongoDB, Express.js, React (Vite), Node.js, Clerk Authentication, Stripe API, Cloudinary, Tailwind CSS, Axios, REST APIs
<https://lms-frontend-two-eosin.vercel.app/>

12. Doctor Appointment Booking System

Full Stack MERN Application

Platform for patients, doctors and admin users. Implemented secure auth, doctor search, appointment scheduling, profile management and admin controls.

- Core Technologies: React, Vite, Tailwind CSS, React Router, Node.js, Express, MongoDB, Cloudinary, REST APIs, Vercel
<https://doctor-appointment-booking-system-r-rosv.vercel.app/>

13. Finance Manager

Finance management web application built using Django

Systematized Finance Manager, a full-featured personal finance management web application using Django. The platform allows users to track income, expenses, budgets, and accounts while providing AI/ML-driven insights such as expense prediction, anomaly detection, and financial recommendations. Includes a RESTful API, interactive charts, and a responsive web interface.

- Tech Stack: Django, HTML, CSS, JavaScript, Plotly.js, PostgreSQL, Scikit-learn, Pandas, NumPy,

14. Car Marketplace

Full Stack Car Marketplace web application | Laravel

Built a car marketplace application with Laravel, featuring secure authentication, a custom drag-and-drop multi-image listing picker, Cloudinary-based cloud image storage, and advanced search and filtering by maker, model, year, price, and fuel type. Included a watchlist system for saving favorite listings.

- Tech Stack: Laravel 11, PHP, Tailwind CSS, Vite, MySQL (Aiven Cloud), Cloudinary

VOLUNTEERING

Active Member | IEEE Student Chapter of SLTC

| 2023 – Present

Leader Board Member – Outing Global Talent | AIESEC in SLTC

| 2024

Project Manager – Aeronautical Division | SEDS in SLTC

| 2024

REFERENCES

Ms. Dilanka de Silva

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